

Experimental Design & Scope of Inference



Observational studies vs. experiments

- 1 A researcher records the exercise habits and cholesterol levels of 500 people without intervening in their lifestyle. Identify whether this is an observational study or an experiment, and explain what type of conclusion (association or causation) it can support.
 - 2 A researcher randomly assigns 200 volunteers to either an exercise program or a no-exercise control group, then measures cholesterol after 6 months. Identify this design, and explain why it can support a causal conclusion.
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The three principles of experimental design (RRCB)

- 3 State the three key principles of a well-designed experiment (sometimes remembered as RRC: Random assignment, Replication, Control of confounding variables), and briefly explain the purpose of each.
 - 4 A study testing a new fertilizer randomly assigns 60 plants to fertilizer or no-fertilizer groups, but only uses 3 plants per group. Identify the design flaw, and explain the consequence.
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Blocking and matched pairs designs

- 5 A study on a new medication expects men and women to respond differently. Describe how a randomized block design would be set up, using sex as the blocking variable.
 - 6 A researcher wants to compare two brands of running shoes on the same 20 runners, having each runner test both brands (in random order) and comparing their times. Identify this design.
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Placebo effect and blinding

- 7 Explain what the placebo effect is, and describe how a single-blind experiment design controls for it.
 - 8 Explain the difference between a single-blind and a double-blind experiment.
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Scope of inference

- 9 A study uses a simple random sample from the population but is purely observational (no random assignment of treatment). State what type of conclusion (about the population, about causation, both, or neither) is justified, and explain why.
- 10 A study randomly assigns treatments to a group of volunteers who were not randomly selected from any larger population. State what type of conclusion is justified.
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Putting it together: evaluating a study design

- 11 A pharmaceutical company recruits 300 volunteers (not randomly selected from the general population) and randomly assigns them to receive either a new drug or a placebo, in a double-blind design. Evaluate what conclusions can and cannot be drawn from this study.
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Confounding variables

- 12 An observational study finds that people who drink more coffee have a higher rate of heart disease. Explain how smoking could act as a confounding variable in this relationship.
- 13 Explain how a well-designed randomized experiment would help rule out a variable like smoking as a confounder in the coffee and heart disease example.
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Identifying experimental units and treatments

- 14 A researcher tests three different fertilizer amounts (low, medium, high) on tomato plants, measuring final yield. Identify the experimental units, the treatment, and the response variable.
- 15 A pharmaceutical researcher is testing whether a new sleep aid reduces the time it takes people to fall asleep, and worries that people with severe insomnia might respond very differently from people with mild insomnia. Design a study using a randomized block design that accounts for this concern, describing each step, and explain the specific benefit this design provides over a simple completely randomized design in this scenario.
- 16 A company tests a new website design by randomly assigning visitors to see either the old design or the new design, without telling visitors which version is the "experimental" one, and measures the percentage who complete a purchase. Explain why this study can support a causal conclusion about the website redesign, and describe the specific scope (which population) that conclusion applies to.

- 17** A university studies whether a new tutoring program improves final exam scores. Volunteers who sign up for tutoring are compared to students who did not sign up, without any random assignment. At the end of the semester, tutored students score noticeably higher on average. Explain why this study cannot establish that the tutoring program caused the improvement, identifying a specific plausible confounding variable.
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Multiple choice — easy 15

- 18** A researcher randomly assigns 467 volunteers to receive a daily supplement or serve as a control group, then measures the outcome after several weeks. Identify the type of study.
- a.** Observational study
 - b.** Census
 - c.** Experiment
 - d.** Meta-analysis
- 19** A researcher records data on exercise habits and cholesterol levels from 435 people without intervening in their lifestyle in any way. Identify the type of study.
- a.** Observational study
 - b.** Experiment
 - c.** Meta-analysis
 - d.** Census
- 20** A researcher randomly assigns 208 volunteers to receive a new exercise program or serve as a control group, then measures the outcome after several weeks. Identify the type of study.
- a.** Observational study
 - b.** Census
 - c.** Experiment
 - d.** Meta-analysis

- 21** A researcher records data on fertilizer use and crop yield from 438 people without intervening in their lifestyle in any way. Identify the type of study.
- a.** Observational study
 - b.** Experiment
 - c.** Census
 - d.** Meta-analysis
- 22** A researcher randomly assigns 259 volunteers to receive a new medication or serve as a control group, then measures the outcome after several weeks. Identify the type of study.
- a.** Census
 - b.** Meta-analysis
 - c.** Observational study
 - d.** Experiment
- 23** A researcher records data on exercise habits and cholesterol levels from 352 people without intervening in their lifestyle in any way. Identify the type of study.
- a.** Meta-analysis
 - b.** Experiment
 - c.** Observational study
 - d.** Census
- 24** A researcher randomly assigns 108 volunteers to receive a meditation app or serve as a control group, then measures the outcome after several weeks. Identify the type of study.
- a.** Experiment
 - b.** Census
 - c.** Meta-analysis
 - d.** Observational study

- 25** A researcher records data on exercise habits and cholesterol levels from 304 people without intervening in their lifestyle in any way. Identify the type of study.
- a.** Experiment
 - b.** Meta-analysis
 - c.** Observational study
 - d.** Census
- 26** A researcher randomly assigns 476 volunteers to receive a tutoring program or serve as a control group, then measures the outcome after several weeks. Identify the type of study.
- a.** Census
 - b.** Meta-analysis
 - c.** Observational study
 - d.** Experiment
- 27** A researcher records data on exercise habits and cholesterol levels from 164 people without intervening in their lifestyle in any way. Identify the type of study.
- a.** Observational study
 - b.** Meta-analysis
 - c.** Census
 - d.** Experiment
- 28** A researcher randomly assigns 224 volunteers to receive a meditation app or serve as a control group, then measures the outcome after several weeks. Identify the type of study.
- a.** Meta-analysis
 - b.** Experiment
 - c.** Census
 - d.** Observational study

- 29** A researcher records data on medication use and symptom severity from 391 people without intervening in their lifestyle in any way. Identify the type of study.
- a.** Observational study
 - b.** Experiment
 - c.** Census
 - d.** Meta-analysis
- 30** A researcher randomly assigns 384 volunteers to receive a new medication or serve as a control group, then measures the outcome after several weeks. Identify the type of study.
- a.** Meta-analysis
 - b.** Census
 - c.** Observational study
 - d.** Experiment
- 31** A researcher records data on meditation habits and stress levels from 301 people without intervening in their lifestyle in any way. Identify the type of study.
- a.** Experiment
 - b.** Observational study
 - c.** Census
 - d.** Meta-analysis
- 32** A researcher randomly assigns 283 volunteers to receive a daily supplement or serve as a control group, then measures the outcome after several weeks. Identify the type of study.
- a.** Observational study
 - b.** Census
 - c.** Meta-analysis
 - d.** Experiment

Multiple choice — medium 15

- 33** In a study on medication use and symptom severity, a simple random sample of 130 people is selected from the population of interest; they are then randomly assigned to receive a new medication or a control condition. What type of conclusion is justified by this design?
- a. Causation, but only about the subjects actually studied (not generalizable)
 - b. Causation, generalizable to the entire population
 - c. Association only, and only about the subjects actually studied (not generalizable)
 - d. Association only, generalizable to the entire population
- 34** In a study on fertilizer use and crop yield, 58 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a new fertilizer or a control condition. What type of conclusion is justified by this design?
- a. Causation, but only about the subjects actually studied (not generalizable)
 - b. Association only, generalizable to the entire population
 - c. Causation, generalizable to the entire population
 - d. Association only, and only about the subjects actually studied (not generalizable)
- 35** In a study on tutoring attendance and exam performance, a simple random sample of 361 people is selected from the population of interest; researchers simply record whether each person already uses a tutoring program or not, without assigning anything. What type of conclusion is justified by this design?
- a. Causation, generalizable to the entire population
 - b. Association only, generalizable to the entire population
 - c. Causation, but only about the subjects actually studied (not generalizable)
 - d. Association only, and only about the subjects actually studied (not generalizable)

- 36** In a study on tutoring attendance and exam performance, 127 volunteers (not randomly selected from any larger population) take part; researchers simply record whether each person already uses a tutoring program or not, without assigning anything. What type of conclusion is justified by this design?
- a.** Association only, generalizable to the entire population
 - b.** Causation, but only about the subjects actually studied (not generalizable)
 - c.** Causation, generalizable to the entire population
 - d.** Association only, and only about the subjects actually studied (not generalizable)
- 37** In a study on tutoring attendance and exam performance, a simple random sample of 275 people is selected from the population of interest; they are then randomly assigned to receive a tutoring program or a control condition. What type of conclusion is justified by this design?
- a.** Causation, generalizable to the entire population
 - b.** Association only, generalizable to the entire population
 - c.** Association only, and only about the subjects actually studied (not generalizable)
 - d.** Causation, but only about the subjects actually studied (not generalizable)
- 38** In a study on tutoring attendance and exam performance, 459 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a tutoring program or a control condition. What type of conclusion is justified by this design?
- a.** Causation, but only about the subjects actually studied (not generalizable)
 - b.** Association only, generalizable to the entire population
 - c.** Causation, generalizable to the entire population
 - d.** Association only, and only about the subjects actually studied (not generalizable)

- 39** In a study on tutoring attendance and exam performance, a simple random sample of 295 people is selected from the population of interest; researchers simply record whether each person already uses a tutoring program or not, without assigning anything. What type of conclusion is justified by this design?
- a.** Causation, but only about the subjects actually studied (not generalizable)
 - b.** Association only, generalizable to the entire population
 - c.** Association only, and only about the subjects actually studied (not generalizable)
 - d.** Causation, generalizable to the entire population
- 40** In a study on tutoring attendance and exam performance, 149 volunteers (not randomly selected from any larger population) take part; researchers simply record whether each person already uses a tutoring program or not, without assigning anything. What type of conclusion is justified by this design?
- a.** Association only, and only about the subjects actually studied (not generalizable)
 - b.** Causation, but only about the subjects actually studied (not generalizable)
 - c.** Association only, generalizable to the entire population
 - d.** Causation, generalizable to the entire population
- 41** In a study on medication use and symptom severity, a simple random sample of 358 people is selected from the population of interest; they are then randomly assigned to receive a new medication or a control condition. What type of conclusion is justified by this design?
- a.** Association only, generalizable to the entire population
 - b.** Causation, generalizable to the entire population
 - c.** Association only, and only about the subjects actually studied (not generalizable)
 - d.** Causation, but only about the subjects actually studied (not generalizable)

- 42** In a study on medication use and symptom severity, 471 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a new medication or a control condition. What type of conclusion is justified by this design?
- a.** Association only, and only about the subjects actually studied (not generalizable)
 - b.** Causation, but only about the subjects actually studied (not generalizable)
 - c.** Causation, generalizable to the entire population
 - d.** Association only, generalizable to the entire population
- 43** In a study on medication use and symptom severity, a simple random sample of 476 people is selected from the population of interest; researchers simply record whether each person already uses a new medication or not, without assigning anything. What type of conclusion is justified by this design?
- a.** Causation, generalizable to the entire population
 - b.** Association only, generalizable to the entire population
 - c.** Association only, and only about the subjects actually studied (not generalizable)
 - d.** Causation, but only about the subjects actually studied (not generalizable)
- 44** In a study on medication use and symptom severity, 165 volunteers (not randomly selected from any larger population) take part; researchers simply record whether each person already uses a new medication or not, without assigning anything. What type of conclusion is justified by this design?
- a.** Causation, generalizable to the entire population
 - b.** Association only, and only about the subjects actually studied (not generalizable)
 - c.** Association only, generalizable to the entire population
 - d.** Causation, but only about the subjects actually studied (not generalizable)

- 45** In a study on supplement use and energy levels, a simple random sample of 406 people is selected from the population of interest; they are then randomly assigned to receive a daily supplement or a control condition. What type of conclusion is justified by this design?
- a.** Association only, and only about the subjects actually studied (not generalizable)
 - b.** Causation, generalizable to the entire population
 - c.** Association only, generalizable to the entire population
 - d.** Causation, but only about the subjects actually studied (not generalizable)
- 46** In a study on supplement use and energy levels, 172 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a daily supplement or a control condition. What type of conclusion is justified by this design?
- a.** Causation, but only about the subjects actually studied (not generalizable)
 - b.** Causation, generalizable to the entire population
 - c.** Association only, generalizable to the entire population
 - d.** Association only, and only about the subjects actually studied (not generalizable)
- 47** In a study on fertilizer use and crop yield, a simple random sample of 192 people is selected from the population of interest; researchers simply record whether each person already uses a new fertilizer or not, without assigning anything. What type of conclusion is justified by this design?
- a.** Causation, generalizable to the entire population
 - b.** Association only, generalizable to the entire population
 - c.** Causation, but only about the subjects actually studied (not generalizable)
 - d.** Association only, and only about the subjects actually studied (not generalizable)

Multiple choice — hard 15

- 48** In a study testing a new medication, subjects know whether they are receiving the real treatment or a placebo, and so do the researchers evaluating the results. Identify the blinding used.
- a.** Not blinded
 - b.** Double-blind
 - c.** Single-blind
 - d.** Triple-blind
- 49** In a study testing a new fertilizer, subjects do not know whether they are receiving the real treatment or a placebo, but the researchers evaluating the results do know which group each subject is in. Identify the blinding used.
- a.** Not blinded
 - b.** Single-blind
 - c.** Triple-blind
 - d.** Double-blind
- 50** In a study testing a new fertilizer, neither the subjects nor the researchers evaluating the results know which subjects received the real treatment versus a placebo. Identify the blinding used.
- a.** Double-blind
 - b.** Triple-blind
 - c.** Single-blind
 - d.** Not blinded
- 51** In a study testing a new exercise program, subjects know whether they are receiving the real treatment or a placebo, and so do the researchers evaluating the results. Identify the blinding used.
- a.** Double-blind
 - b.** Single-blind
 - c.** Not blinded
 - d.** Triple-blind

- 52** In a study testing a tutoring program, subjects do not know whether they are receiving the real treatment or a placebo, but the researchers evaluating the results do know which group each subject is in. Identify the blinding used.
- a.** Single-blind
 - b.** Not blinded
 - c.** Triple-blind
 - d.** Double-blind
- 53** A study on a new fertilizer randomly assigns 132 subjects directly to treatment or control groups, with no grouping by any other characteristic beforehand. Identify this experimental design.
- a.** Completely randomized design
 - b.** Randomized block design
 - c.** Matched pairs design
 - d.** Stratified design
- 54** A study on a new exercise program first separates subjects into groups based on sex, then randomly assigns subjects to treatment or control within each group separately. Identify this experimental design.
- a.** Randomized block design
 - b.** Matched pairs design
 - c.** Completely randomized design
 - d.** Stratified design

- 55** A study on a new fertilizer has each of 37 subjects try both the treatment and the control condition (in random order), comparing each subject's two results directly. Identify this experimental design.
- a.** Matched pairs design
 - b.** Completely randomized design
 - c.** Randomized block design
 - d.** Stratified design
- 56** A study on a meditation app randomly assigns 83 subjects directly to treatment or control groups, with no grouping by any other characteristic beforehand. Identify this experimental design.
- a.** Randomized block design
 - b.** Stratified design
 - c.** Completely randomized design
 - d.** Matched pairs design
- 57** A study on a new fertilizer first separates subjects into groups based on age group, then randomly assigns subjects to treatment or control within each group separately. Identify this experimental design.
- a.** Stratified design
 - b.** Randomized block design
 - c.** Matched pairs design
 - d.** Completely randomized design
- 58** A study on a new exercise program randomly assigns subjects to treatment or control, but uses only 3 subjects per group. Identify the main design flaw.
- a.** Lack of random assignment (confounding is likely)
 - b.** Lack of a control group (no baseline for comparison)
 - c.** Lack of replication (too few subjects per group)
 - d.** Lack of blinding (placebo effect not controlled)

- 59** A study on a new medication compares volunteers who chose to use it to volunteers who chose not to, without randomly assigning anyone to either group. Identify the main design flaw.
- a.** Lack of replication (too few subjects per group)
 - b.** Lack of random assignment (confounding is likely)
 - c.** Lack of a control group (no baseline for comparison)
 - d.** Lack of blinding (placebo effect not controlled)
- 60** A study on a tutoring program tells subjects exactly which group (treatment or control) they are in, with no placebo used. Identify the main design flaw.
- a.** Lack of a control group (no baseline for comparison)
 - b.** Lack of blinding (placebo effect not controlled)
 - c.** Lack of random assignment (confounding is likely)
 - d.** Lack of replication (too few subjects per group)
- 61** A study on a tutoring program gives every single subject the treatment, with no separate group left untreated for comparison. Identify the main design flaw.
- a.** Lack of a control group (no baseline for comparison)
 - b.** Lack of random assignment (confounding is likely)
 - c.** Lack of replication (too few subjects per group)
 - d.** Lack of blinding (placebo effect not controlled)
- 62** A study on a tutoring program randomly assigns subjects to treatment or control, but uses only 2 subjects per group. Identify the main design flaw.
- a.** Lack of a control group (no baseline for comparison)
 - b.** Lack of blinding (placebo effect not controlled)
 - c.** Lack of replication (too few subjects per group)
 - d.** Lack of random assignment (confounding is likely)